MATCHMINDAI - PROJECT PRESENTATION DECK

Real-Time Sports Intelligence & 10,000-Run Monte Carlo Match Prediction Engine

Live Web Application: matchmindai.streamlit.app	**GitHub Repository:** github.com/Soham-ss/MatchMindAI
Core Stack: Python 3.14, Streamlit, Scikit-Learn, Plotly	**Engine:** 10,000 Monte Carlo Runs + ML RandomForest

■ Slide 2: The Problems We Solved (Why We Built MatchMindAi)

■ **Problem 1: Static Historical Data Only**	Traditional prediction websites only look at past matches from years ago. They completely ignore today's live pitch reports, weather changes, player injuries, or stadium dew factors.
■ **Problem 2: Messy AI Data Dumps**	General AI chatbots return long, messy text links (e.g. Source [1]...) instead of simple, crisp 1-line direct answers.
■ **Problem 3: Lack of Match Simulation**	Cricket fans and fantasy players cannot simulate thousands of ball-by-ball match scenarios to view score distribution ranges.

■ Slide 3: Our Solution (How MatchMindAi Fixes Everything)

■ **10,000 Monte Carlo Runs**	Simulates 10,000 ball-by-ball matches per click based on team strength ratings, venue pitch profiles, and stadium dew factors.
■ **1-Line Factual Answers**	Instantly answers player team queries, Orange Cap winners, Purple Cap winners, and Champions for any year (2008–2026).
■ **Live Web Intelligence**	Scrapes live Google News RSS feeds to incorporate today's pitch, weather, and team squad updates.
■ **High-Energy Cyber UI**	Dark cyberpunk theme, live stadium video header, cursor depth particle radar, and sticky bottom Chat Box.

■ Slide 4: Main Platform Features & Highlights

	Recognizes user name introductions personally (e.g. *'Hi my name is Soham'* ➔ *'Hello Soham! ■'*).	**■ 18-Year IPL Database**	Maps Davic
	Detects invalid pre-2008 queries (*'who won in 2005'*) explaining IPL started in 2008 with Shaun Marsh.	**■ 1-Click Match Predictor**	Pick histog

■■ Slide 5: System Architecture & Workflow (How Data Flows)

Step 1: User Input	User inputs a question or selects 2 teams & stadium venue.
Step 2: Intent Router	Classifies query into Greeting, Bot Identity, IPL History Lookup, or Match Prediction.
Step 3: Web Scraper Agent	Scrapes live Google News RSS feeds for real-time weather & squad updates.
Step 4: Monte Carlo Agent	Executes 10,000 ball-by-ball stochastic match outcome simulations.
Step 5: ML RandomForest Agent	Queries pre-trained machine learning model on historical IPL match features.
Step 6: Blended Ensemble	Combines 55% Monte Carlo + 45% ML to compute final win probabilities and report.

■ Slide 6: Machine Learning & Simulation Math Explained Simply

■ **RandomForest Classifier**	Trained on historical IPL datasets using scikit-learn. Encodes teams, venue, toss winner, and toss decision. Pre-compiled weights are saved as .pkl binary files in models/.
■ **Monte Carlo Math**	Calculates run probabilities per ball based on team rating (CSK 88, MI 87, RCB 85) and venue dew factor (Wankhede 0.12). Runs 10,000 iterations in milliseconds.
■ **Score Distributions**	Generates interactive Plotly bar charts showing expected score ranges and average totals for both competing teams.

■ Slide 7: Technical Challenges Faced & Solved

■ **GitHub 100 MB Limit Error**	■ **Solution:** Removed IPL.csv (104 MB) from git history because model weights were already compiled into .pkl binary files in models/.
■ **Streamlit Cloud Path Errors**	■ **Solution:** Replaced relative string paths with Python's os.path.abspath to dynamically locate exact folder paths on Linux.
■ **Missing Cloud Packages**	■ **Solution:** Added beautifulsoup4 and python-dotenv to requirements.txt and wrapped imports in safe try...except blocks.
■ **Phrasing Variations**	■ **Solution:** Expanded regex keywords to catch all tense variations ('who wins', 'wins', 'win', 'won').

Slide 8: UI/UX & High-Energy Design System

Cyberpunk Dark Aesthetic	Designed with a sleek #030712 background and HSL tailored emerald accents (#10b981, #34d399).
Modern Typography	Uses Google Font (Outfit) for maximum legibility across laptops, tablets, and mobile devices.
Dynamic Visual FX	Features a looping HD stadium lights video banner and an interactive 4D particle cursor depth radar.
Sticky Bottom Chat Input	Pinned Chat Box stays permanently visible at the bottom of the viewport when scrolling long chat histories.

■ Slide 9: Cloud Deployment & Infrastructure

■■ Live Hosting	Hosted live on **Streamlit Community Cloud** with 100% uptime. Accessible globally at matchmindai.streamlit.app.
■ Source Control	Version-controlled on **GitHub** (github.com/Soham-ss/MatchMindAI).
■ Automated CI/CD	Pushing code updates to GitHub main branch automatically triggers a rebuild on Streamlit Cloud within ~30 seconds.

■ Slide 10: Conclusion & Future Roadmap

■ **Conclusion**	**MatchMindAi** successfully combines Artificial Intelligence, Machine Learning, Monte Carlo Simulations, and Real-Time Internet Scraping into a fast, accurate, and visually stunning web platform.
■ **Live Score API Integration**	Connect live cricket APIs for real-time win probability shifts during active live matches.
■ **Fantasy Team Optimizer**	Auto-generate optimal Dream11 / My11Circle lineups based on Monte Carlo expected points.
■ **Multi-Sport Expansion**	Extend simulation models to Football (Premier League, Champions League) and Basketball (NBA).